

Article Critique #3

Faruq Pradhan MBBCh

For my third article critique, I chose the publication “Palliative Care Always: Hepatology-Virtual Primary Palliative Care Training for Hepatologists” by Denofrio et al. (2022), published in *Hepatology Communications*. This study describes the development and implementation of an online course “Palliative Care Always: Hepatology” (PCA:Hep) designed to train hepatologists in palliative care medicine skill specifically for patients with end-stage liver disease. PCA:Hep was embedded within a larger PAL-LIVER PCORI clinical trial which compared palliative care delivery between palliative specialists and hepatologists that received additional palliative education. PCA:Hep represents the implementation of the palliative care skills training provided to the hepatologist-arm of PAL-LIVER. The 12-week online course was hosted on Stanford University’s edX platform with 11 modules developed. Formative evaluation assessed four domains: feasibility, knowledge acquisition, self-efficacy, and practice patterns. 39 hepatology providers across 10 clinical sites completed the course. Given the increased mortality rate known to exist for patients with end-stage liver disease (including both transplant and non-transplant candidates), this study addresses an important gap in specialist training especially as guidelines support early integration into patient care.

A strength of the study is the background and clear definition of the underlying problem being addressed by the study; namely, the high symptom burden and mortality associated with end-stage liver disease and underutilization of palliative care. Appropriate

references are made with relation to the limited workforce for palliative care, and discusses why hepatologists, as the primary providers for much of the care of these patients, are well-positioned to help as a part of the solution. The logical extension from clinical needs to educational intervention is well discussed and adds to the strength of the study design overall. While the study design itself is housed within a larger research context (i.e., within LIVER-PAL), the authors did a good job in clearly describing how PCA:Hep fits.

Another strength of the study is the multimodal approach used to assess educational effectiveness. Instead of relying on just one assessment tool (such as pre/post testing), the authors combined multiple different tools (i.e., feasibility assessment, pre/post multiple-choice testing, self-assessment through pre/post-course Likert surveys, longitudinal practice pattern changes assessed at multiple time points). This approach allowed for long-term educational effectiveness evaluation given the practice pattern changes evaluated at up to 2 years. A common critique of similar studies that integrate eLearning is the lack of long-term assessment or evaluation of long-term impact on practice patterns. As a strength, this study was well designed to address these critiques through longitudinal practice pattern assessment at 6-, 12-, and 24-month intervals. Over 80% of hepatologists reporting integration of palliative care skills into routine clinical practice at the 1-year mark is a particularly notable outcome that speaks to the strength of study design and presents meaningful evidence of real-world impact.

The statistical methodology appropriately chosen per assessment tool is another strength of the study. For the non-parametric paired data collected via pre- and post-knowledge assessment, paired Wilcoxon signed-rank test is logical. The authors also

appropriately reflect on the challenges with the two-sample Wilcoxon test used for self-efficacy analysis given the anonymous nature of the surveys (and hence an inability to account for direct correlation between individual self-efficacy assessments) and impact on the p-value. The transparency of this is key for reproducibility purposes, and for reader understanding.

One weakness of the study is related to the pretest and final exam being identical 20 multiple-choice questions and a lack of details regarding item development, or prior validation of the questions. As one of the primary outcomes of the study, along with the high pretest mean score of 85.9% despite the majority of participants lacking prior formal palliative care training, raises red flags regarding the sensitivity of the questions to adequately assess baseline knowledge. The final exam results showed a 6% score improvement to 91.3% compared to pre-test scoring, with a confidence interval of 2.8-9.2 and $p = 0.001$. Though reaching statistical significance, a 6% improvement on a 20-item test represents scoring 1 additional question correctly. The clinical relevance of such a small improvement is not discussed by the authors.

Another weakness of the study is the lack of educational theory foundation discussion in the study design methodology. Though there is an appreciated number of details regarding course content, the grounding of pedagogical design choices based on specific educational theories is not discussed. Though references and comparisons are made to other peer-reviewed studies related to palliative care training, connections to educational theory frameworks such as Kolb's experiential learning theory, or Bandura's

social cognitive theory would have meaningfully improved the study and strengthened the conclusions derived.

A third weakness of the study relates to the self-selected nature of participants and the lack of a comparison group. The 39 participants were a self-selected cohort within the context of the larger PAL-LIVER trial which introduces bias as these are likely clinicians who already recognized the importance of palliative care in patients with liver disease and could have created skewed results. This was appropriately mentioned in the discussion section by the authors and could have contributed towards higher pretest scores; however, as noted earlier, the lack of details regarding the 20-question instrument further compounds this study weakness. Furthermore, the lack of a power analysis is notable given the small size of the study population and the relatively small change in pre/posttest scoring.

Reference:

DeNofrio, J. C., Verma, M., Kosinski, A. S., Navarro, V., Taddei, T. H., Volk, M. L., Bakitas, M., & Ramchandran, K. (2022). Palliative care always: Hepatology — Virtual primary palliative care training for hepatologists. *Hepatology Communications*, 6(4), 920–930. <https://doi.org/10.1002/hep4.1849>